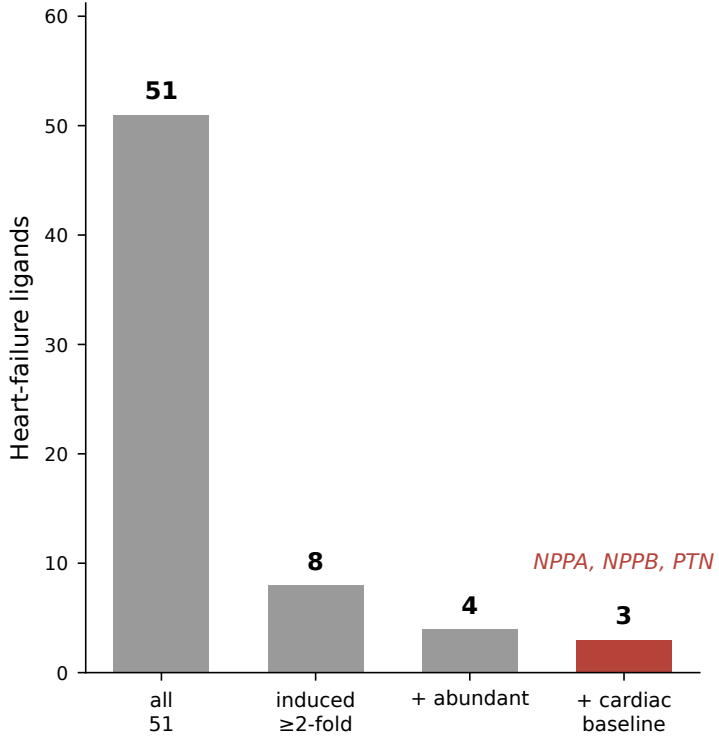
Does the failing heart send a signal that any tumor is equipped to receive?

A multi-agent pipeline of six deterministic skills screened 188 ligand-receptor interactions against 9,538 primary tumors across 30 types.

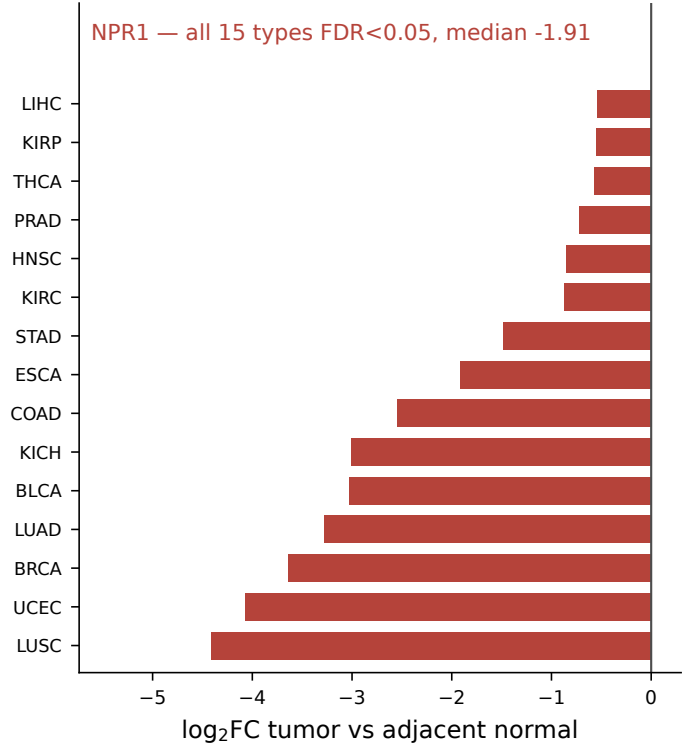
a

Only 3 ligands are distinctively cardiac



b

Its receptor is depleted, not enriched



c

0

of 146 axes pair a cardiac-distinctive ligand with a tumor-enriched receptor

The guardrails are the transferable result

- **Absolute expression floor**
  without it, random receptor sets reproduce the ranking (ρ 0.363 → 0.107)
- **Expression-matched null**
  4 of 30 types beat it; a naive pool inflates the apparent signal
- **Assertions, not advice**
  requirements halt execution rather than sit in a methods section